Jacobian of Alpoge's candidate counterexample map $F : \mathbb{C}^3 \to \mathbb{C}^3$

$$F(x,y,z) \;=\; \Big((1+xy)^3 z + y^2(1+xy)(4+3xy), \quad y + 3x(1+xy)^2 z + 3xy^2(4+3xy), \quad 2x - 3x^2 y - x^3 z \Big)$$

$$\frac{\partial F}{\partial(x,y,z)} \;=\; \begin{pmatrix} 3y^3\left(xy+1\right) + y^3\left(3xy+4\right) + 3yz\left(xy+1\right)^2 & 3xy^2\left(xy+1\right) + xy^2\left(3xy+4\right) + 3xz\left(xy+1\right)^2 + 2y\left(xy+1\right)\left(3xy+4\right) & \left(xy+1\right)^3 \\ 9xy^3 + 6xyz\left(xy+1\right) + 3y^2\left(3xy+4\right) + 3z\left(xy+1\right)^2 & 9x^2y^2 + 6x^2z\left(xy+1\right) + 6xy\left(3xy+4\right) + 1 & 3x\left(xy+1\right)^2 \\ -3x^2z - 6xy + 2 & -3x^2 & -x^3 \end{pmatrix}$$

$$\det\!\left(\frac{\partial F}{\partial(x,y,z)}\right) \;=\; -2 \qquad \text{(SymPy-verified identity in } \mathbb{Q}[x,y,z])$$

Three distinct points collapse to the same image:

$$F\big(0,\,0,\,-\tfrac{1}{4}\big) \;=\; F\big(1,\,-\tfrac{3}{2},\,\tfrac{13}{2}\big) \;=\; F\big(-1,\,\tfrac{3}{2},\,\tfrac{13}{2}\big) \;=\; \big(-\tfrac{1}{4},\,0,\,0\big)$$

At each small-integer point (x,y,z) below, the middle column shows $J(x,y,z)$ as a 3×3 integer matrix, and the right column is its determinant. Any row is human-verifiable: pick a point, substitute into the formulas for $\partial f_i/\partial(x,y,z)$, and take the 3×3 determinant (Sarrus is easiest here). Fewer points than the MC step because the full matrices need room — the script's rigorous MC uses $[-1000,1000]^3$, $k{=}30$, for a Schwartz–Zippel false-positive bound $\lesssim 2.2\times 10^{-65}$.

point	J	det	point	J	det	point	J	det
$(0,0,0)$	$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}$	-2	$(0,0,1)$	$\begin{pmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}$	-2	$(1,-1,1)$	$\begin{pmatrix} -1 & 1 & 0 \\ -6 & 4 & 0 \\ 5 & -3 & -1 \end{pmatrix}$	-2
$(1,0,0)$	$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 3 \\ 2 & -3 & -1 \end{pmatrix}$	-2	$(1,1,0)$	$\begin{pmatrix} 13 & 41 & 8 \\ 30 & 52 & 12 \\ -4 & -3 & -1 \end{pmatrix}$	-2	$(2,1,-1)$	$\begin{pmatrix} -8 & 44 & 27 \\ -15 & 85 & 54 \\ 2 & -12 & -8 \end{pmatrix}$	-2
$(0,1,0)$	$\begin{pmatrix} 7 & 8 & 1 \\ 12 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}$	-2	$(-1,1,0)$	$\begin{pmatrix} 1 & -1 & 0 \\ -6 & 4 & 0 \\ 8 & -3 & 1 \end{pmatrix}$	-2	$(-1,2,1)$	$\begin{pmatrix} -34 & 25 & -1 \\ -81 & 55 & -3 \\ 11 & -3 & 1 \end{pmatrix}$	-2

Map and points due to Levent Alpoge (`@__alpoge__`), announcement 2026-07-20.
This document is a symbolic re-derivation only.